Multi-Algorithmic Simulation of Derivative Structures and Valuation of Index Options

GitHub Project Repository

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Open Source Project

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Objective

Develop and implement a Python-based quantitative pricing Simulation for **valuing Call and Put Options**. The project applies **numerical algorithms** (Monte Carlo), to **evaluate and compare** the characteristics of **different option styles** (American, European, Asian) across major **global equity indices** ((DAX, S&P 500, and Nikkei 225)).

- **European Option: DAX Index (Exercise on expiration)**
- **American Option: S&P 500 Index (Early Exercise Premium)**
- **Asian Option: Nikkei 225 Index (Path dependency)**

Goals

1. Estimate **historical volatility** in real time from market data (Last 2 Years).
2. Determine **fair option prices** under no-arbitrage conditions.
3. Compute the **probability of profit (PoP)** for each option type.
4. Compare European, American, and Asian **option structures** in terms of **flexibility**, **volatility sensitivity**, and **probability of profit** across the selected indices.

1 Official SEC Definition

- **Option: Contracts** that give the purchaser the **right**, but not the **obligation**, to buy (Call) or sell (Put) a **security** (Stock, Index, fund), at a **fixed price** within a **specific time period.**
- **Call Option:** The right to buy the underlying asset at a fixed price within a specific time period.
- **Put Option:** The right to sell the underlying asset at a fixed price within a specific time period.
- **European Option:** Grants the holder the right to purchase the underlying asset at a fixed strike price (K) only on the expiration date (T).
- **American Option:** Allows exercise at any point up to maturity, offering distinct advantages for dividend capture strategies.
- **Asian Option:** A path-dependent option whose payoff depends on the average price of the underlying asset over a specified period rather than the price at maturity alone.

[1]

[1] **Source:** U.S. Securities and Exchange Commission (SEC), *Investor.gov – Options Glossary.*

2 Methodology: Numerical Option Pricing Framework

This project implements three distinct numerical pricing methodologies. Each method is based on the same stochastic price dynamics but differs in payoff structure and numerical treatment:

- Monte Carlo simulation for **European options**
- Path-dependent Monte Carlo for **Asian options**
- Implicit Finite Difference Method (FDM) for **American options**

2.1 Key Advantages of Monte Carlo

Preferred over analytical models (such as Black-Scholes) in three scenarios:

1. **Path Dependency**
   Analytical models often look only at the starting and end points. Monte Carlo simulates the entire price trajectory, making it essential for instruments like *Asian Options*.
2. **Complex Payoff Rules**
   It adeptly handles exotic options with conditional execution, such as *Barrier Options* (active only if price hits X or stays below Y). Deriving closed-form mathematical formulas for these is often impossible.
3. **High Dimensionality**
   Monte Carlo simulations efficiently manage multi-variable environments *(Basket Options)* without exponential complexity.

3 Stochastic Process: Geometric Brownian Motion

We assume the underlying asset follows a **Geometric Brownian Motion (GBM)** under the risk-neutral measure, where r denotes the risk-free rate, q the dividend yield, σ the volatility, and W_t a Wiener process (Brownian motion).

The continuous-time stochastic differential equation (SDE) is

$$dS_t = (r - q)S_t\,dt + \sigma S_t\,dW_t. \tag{1}$$

Under the risk-neutral measure, the expected return μ is replaced by $(r - q)$. Applying **Itô's Lemma** yields the closed-form solution, which we discretize over small time steps Δt for numerical simulation. The implementation in Python uses

$$S_{t+\Delta t} = S_t \exp\left(\left(r - q - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\,Z\right), \tag{2}$$

Where Z is a random draw from the standard normal distribution: $Z \sim \mathcal{N}(0, 1)$.

Core Simulation Code

```
drift = (r - 0.5 * sigma ** 2) * dt
diffusion = sigma * np.sqrt(dt) * Z[:, t - 1]
S[:, t] = S[:, t - 1] * np.exp(drift + diffusion)
```

4 European Option Pricing via Monte Carlo

4.1 Payoff Function

The value of a call option at maturity (T) depends on where the stock price (S_T) finishes relative to the strike price (K). The payoff formula is:

$$C_T = \max(S_T - K, 0)/P_T = \max(K - S_T, 0) \tag{3}$$

- If $S_T > K$: The Call option is *In-The-Money* (Profit $= S_T - K$).
- If $S_T \leq K$: The Call option is *Out-Of-The-Money* (Worthless, Payoff $= 0$).

4.2 Valuation Formula

The simulation performs the following steps 10,000 times.

1. Generate a random price path from $t = 0$ to $t = T$ and calculate the price S_T at maturity.
2. Calculate the payoff $\max(S_T - K, 0)$.

Monte Carlo approximation:

$$V_0 \approx e^{-rT} \frac{1}{N} \sum_{i=1}^{N} \text{Payoff}_i \tag{4}$$

5 Asian Options: Path-Dependent Monte Carlo

Asian options are path-dependent derivatives where the payoff depends on the average price, A, of the underlying asset over a specified period. This average A can be computed continuously using an arithmetic mean. Asian options are typically cheaper than European options with identical parameters. They are useful for protecting against price manipulation or extreme, short-term price spikes near maturity.

The payoffs at maturity T for a call (C_T) and a put (P_T) with strike K are given by:

$$C_T = \max(A - K, 0), \quad P_T = \max(K - A, 0) \tag{5}$$

The initial value V_0 under the risk-neutral measure Q is the discounted expected payoff:

$$V_0 = e^{-rT} E^Q[\text{Payoff}_T(A)] \tag{6}$$

6 American Options: Implicit Finite Difference Method

American options allow early exercise. Therefore, Monte Carlo alone cannot capture optimal stopping efficiently. Instead, we solve the Black-Scholes Partial Differential Equation (PDE):

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r - q)S\frac{\partial V}{\partial S} - rV = 0 \tag{7}$$

subject to the early-exercise constraint:

$$V(S, t) \geq \Phi(S) \tag{8}$$

where $\Phi(S)$ is the intrinsic value.

6.1 Implicit Finite Difference Scheme

We discretize the Black–Scholes PDE using a fully implicit backward time-stepping scheme. At each time step we solve:

$$AV^n = V^{n+1}, \tag{9}$$

where A is a tridiagonal matrix from the spatial discretization. This requires solving one tridiagonal linear system per time step. These correspond to the lower, main, and upper diagonals of the tridiagonal matrix.

Matrix Coefficients

For grid index j, the coefficients of A are

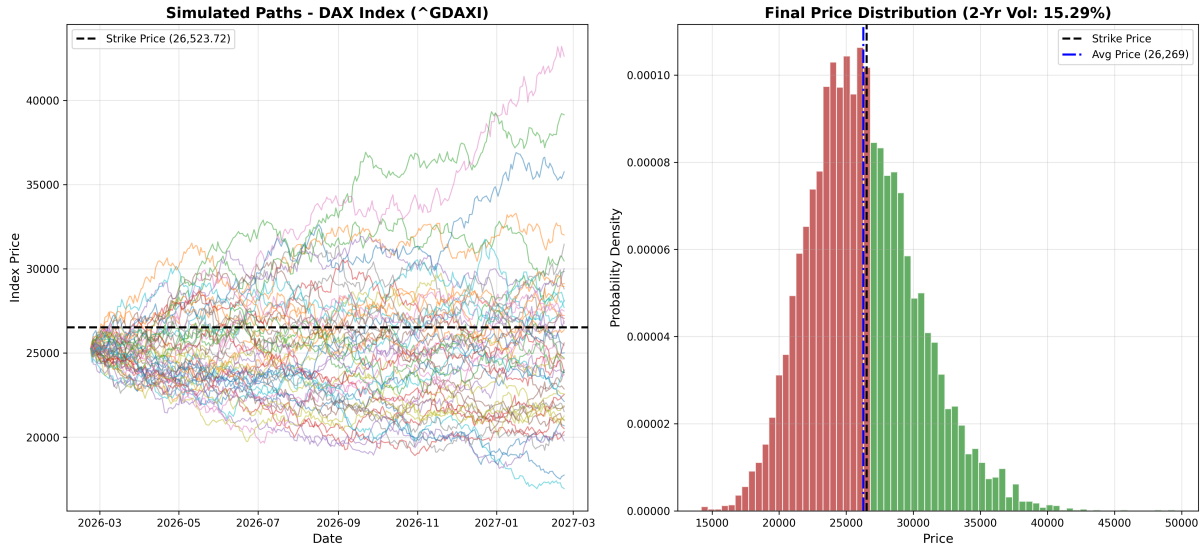
```
alpha = 0.5 * dt * (sigma**2 * j**2 - (r - q) * j)
beta  = 1 + dt * (sigma**2 * j**2 + r)
gamma = 0.5 * dt * (sigma**2 * j**2 + (r - q) * j)
```

After solving for the continuation value, we enforce the American constraint by projecting onto the intrinsic value. This projection step ensures that the option value is never below its immediate exercise value and therefore captures the American early-exercise premium.

$$V = \max(V_{\text{hold}}, \Phi(S)). \tag{10}$$

7 Results

This section presents the Monte Carlo pricing results for 5% out-of-the-money (OTM) call options on the S&P 500 (American), Nikkei 225 (Asian), and DAX Index (European). The figure shows price paths, the terminal price distribution, and key pricing metrics. (Refer to the GitHub Link to see the detailed Results of American and Asian Option.




```
SIMULATION RESULTS: EUROPEAN CALL OPTION PREVIEW
--------------------------------------------------
Asset:                  DAX Index (^GDAXI)
Current Price (S0):     25,260.69
Target Strike (K):      26,523.72 (+5% OTM)
2-Year Volatility:      15.29%
Fair Option Price:      1,430.41 (Using Standard Monte Carlo)
--------------------------------------------------
Probability of Profit: 43.9%
```

Figure 1: European Call Option on the DAX Index.

The DAX index has volatility similar to the S&P 500 (15.3%). The European call premium is the highest relative to spot (5.66%), with a probability of profit of 43.9%. The terminal distribution shows moderate dispersion with a right tail driven by compounding effects.

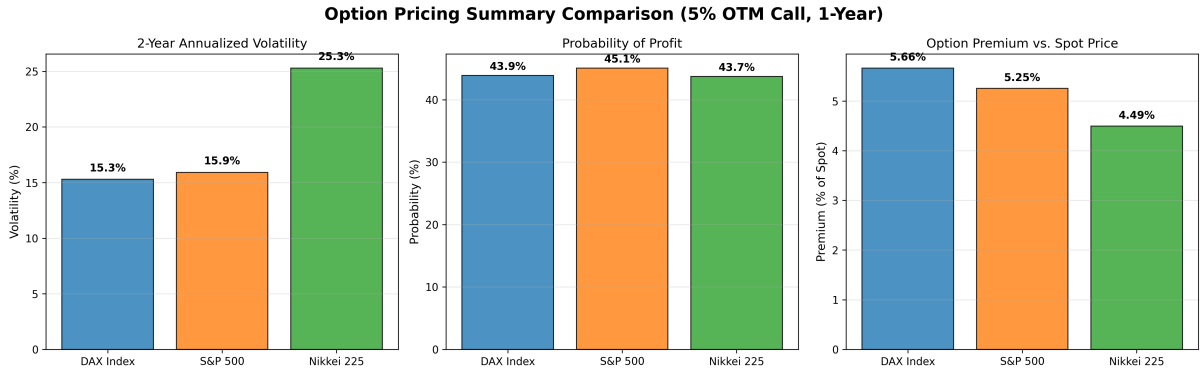


Figure 2: Comparison of Volatility, Probability of Profit, and Premium.

Overall, higher volatility increases dispersion but does not necessarily imply a higher probability of profit. The American option on the S&P 500 provides the highest probability of profit, while the DAX exhibits the highest premium relative to spot. The Asian option shows reduced payoff sensitivity due to averaging, despite elevated volatility.

8 Important Modeling Assumptions

Assume GBM dynamics with constant volatility and risk-free rate, no transaction costs, and continuous trading. A dividend yield q is included in FDM to permit early exercise; dividend assumptions must be consistent across Monte Carlo and FDM implementations.